

MaDIq Labs Omni-Channel Analytics Dashboard

Technical Design, Data Transformation & Power BI Implementation

This document explains the technical workflow behind the dashboard: source ingestion, Power Query transformation, model design, DAX measures, filter context, time intelligence, validation and visualization mapping.

1. End-to-End Architecture

CSV source data → Power Query ingestion → cleaning and data-type enforcement → standardized fields → dimensional model → DAX measure layer → time/filter logic → interactive visuals → business insights.

2. Data Areas

Area	Analytical Purpose
Customer Calls	Voice activity, language, region, outcomes, duration and conversion
SalesIQ AI Interactions	Suggestions, objections, sentiment, segments, confidence and conversion status
AI Performance Logs	Accuracy, confidence, compliance, response time, uptime and processing volume
CRM Updates	Pipeline/deal and commercial context
Automation Events	Automation execution information
Field Sales	Sales/field activity context
Dim Date / Product / Region	Reusable filtering and cross-report analysis

3. Power Query Transformation

- Load CSV sources and promote the first row to headers.
- Apply explicit types: text for identifiers/categories, whole numbers for counts, decimal numbers for continuous metrics and date/datetime for time fields.
- Standardize field names and business terminology.
- Prepare dates for month, quarter and time-intelligence analysis.
- Replace public-facing product values with fictional portfolio names VoxIQ and SalesIQ AI.
- Keep repeatable transformations upstream so every visual consumes consistent data.

4. Data Model

Shared dimensions such as Dim Date, Dim Product and Dim Region provide consistent filtering across operational tables. Domain-specific tables retain detailed records while measures aggregate them into reporting KPIs. Helper/disconnected tables are used where a visual requires a custom metric axis or combines measures from different business domains.

5. DAX Measure Layer

Group	Examples
Engagement	Total Engagements, Engagement Rate, Engagement Growth
Commercial	Revenue Pipeline, Conversion Rate
Voice	Voice Calls, Connected Calls, Call Success Rate, Avg Talk Time, Voice to Conversion
SalesIQ AI	Total Interactions, Suggestions Used, Adoption Rate, Objection Assist, Avg Session Duration
AI Health	AI Accuracy, AI Performance, Confidence, Compliance, Response Time, System Uptime
Time Intelligence	Current period, previous month and MoM variance
Narrative	Dynamic Key Insights based on filter context

6. Month-over-Month Calculation Pattern

KPI cards evaluate the current value and a previous-month value through the date dimension. The difference or percentage change is then formatted into positive, negative or neutral context. This adds direction to the KPI rather than showing only the current number.

7. Product Performance Radar

The radar uses a metric selector/helper table and product-specific measures. SWITCH logic returns the appropriate business measure for the selected metric. Measures with different natural scales are normalized where required. Product-specific measures can return BLANK for the non-selected product so the visual responds correctly to the product slicer.

8. Product Engagement Share

A helper Product Engagement table provides the VoxIQ and SalesIQ AI categories. A SWITCH-based measure maps VoxIQ to the voice activity measure and SalesIQ AI to the interaction activity measure, allowing two different source measures to participate in one share visual.

9. Dynamic Key Insights

Key Insights is measure-driven rather than static text. DAX reads the current filter context, formats KPI values and concatenates them into readable statements. This keeps narrative text aligned with slicer selections and the current dashboard state.

10. Visualization Mapping

Visual	Purpose
KPI Cards	Current value plus MoM direction
Line Chart	Time trends and product comparison
Bar/Column	Magnitude comparison
Donut	Composition/share for limited categories
Heatmap	Day/time intensity

Treemap	Segment composition
Radar	Multi-dimensional product comparison
Scatter	Response-time versus accuracy relationship
Combo Chart	Volume plus rate/uptime on separate scales
Dynamic Text	Business-readable narrative from measures

11. Slicer & Filter Design

Executive Overview uses Product, Region, Quarter and Year & Month. VoxIQ uses Region, Language, Year & Month and Call Outcome. SalesIQ AI uses Region, Sales Rep, Year & Month and Conversation Status. AI Performance uses Product, Accuracy Band, Year & Month and Response Band. Measures recalculate under the active filter context.

12. Data Validation

- Validate source row counts and category counts after transformation.
- Reconcile major KPI totals with source-level aggregations.
- Test month-boundary behaviour in previous-month calculations.
- Verify slicer-to-visual interactions.
- Check hard-coded DAX strings after renaming categories to avoid BLANK outputs.
- Validate percentage scaling and formatting, e.g. 0.8927 versus 89.27.
- Check nulls, data-type conversions and unexpected categories before publishing.

13. Refresh / Deployment

The development model uses local CSV File.Contents sources. Power BI Service cannot directly access a developer's local drive without an appropriate gateway. For a portfolio deployment, cloud-hosted source files such as OneDrive/SharePoint or a configured gateway provide a cleaner refresh architecture.

14. Skills Demonstrated

Power BI • Power Query • DAX • Dimensional Modelling • KPI Design • Time Intelligence • Filter Context • Data Transformation • Data Validation • Interactive Visualization • Dashboard UX • Business Storytelling

15. Portfolio Positioning

MaDIq Labs, VoxIQ and SalesIQ AI are fictional portfolio branding. The public dashboard should be described as a synthetic/demo case study created to demonstrate BI development and analytical thinking.